

Apratim Das

AI & Systems Developer | Multidisciplinary Engineer | Technical Project Lead

B.Tech Electrical & Electronics Engineering, IIT Dharwad (Class of 2028) — 5th Semester

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apratim-portfolio.pages.dev | Agartala, Tripura, India



EEE undergrad who builds across AI/ML, full-stack web, and robotics – shipping working projects fast by pairing LLM-assisted coding with a solid enough grip on Python, JS, and systems fundamentals to read, debug, and extend what the AI writes. Also leads teams and runs large student events, so equally comfortable on the technical or the people side of a project.

Technical Strengths

- End-to-end AI/ML & full-stack system building, from backend architecture to shipped UI
- Rapid prototyping with AI-assisted coding, backed by hands-on debugging across the stack
- Robotics, simulation, and applied ML exposure (ROS2, SLAM, power-systems modelling)

Leadership & Management Strengths

- Led 6+ technical and creative teams across hackathons, competitions, and student bodies
- Owned end-to-end execution of institute-scale events – planning, vendor/merch coordination, timelines
- Coordinated cross-functional groups spanning design, marketing, and engineering under deadline pressure
- Represented IIT Dharwad externally as Campus Ambassador and cultural-meet contingent delegate

Education

Indian Institute of Technology (IIT) Dharwad

Aug 2024 – Aug 2028

B.Tech, Electrical & Electronics Engineering (EEE) | Currently in 5th Semester

Technical Projects

ATLAS — Autonomous Multi-Agent AI Coding Platform

[github repo](#)

Next.js 15, FastAPI, LangGraph, Qdrant, Docker, PostgreSQL, Redis, Celery

- Architected a six-agent pipeline (Planner, Repo Analyst, Coder, Executor, Debugger, Reviewer) orchestrated with **LangGraph** to autonomously plan, generate, run, and self-correct code inside isolated **Docker** sandboxes.
- Built a repository-intelligence layer using **vector embeddings (Qdrant)** for semantic, cross-file code retrieval over large codebases.
- Implemented real-time **WebSocket** streaming of agent reasoning, execution logs, and diff review inside a Next.js + Monaco Editor dashboard; heavy iterative debugging of async agent/task failures.

Nasiko — AI Agent Orchestration Platform (Team Lead, Nasiko AI Buildathon)

[github repo](#)

FastAPI, LangChain, Kong API Gateway, Docker, MongoDB, Redis

- Led **Team Mauryan Minds**; designed an AI agent registry with **LangChain**-driven query routing and confidence-scored agent selection.
- Directed the system architecture across microservices (auth, router, registry, chat-history) unified behind a **Kong API Gateway**, coordinating parallel work across teammates under hackathon time pressure.

SOI Cartographer — Multi-Robot Autonomous Exploration (Competition Solo Entry)

[github repo](#)

ROS2 Jazzy, Gazebo Harmonic, Python, A*, ICP

- Built a cooperative multi-robot **SLAM & exploration** stack – frontier detection with information-gain selection, occupancy-grid mapping, **ICP** map alignment/merging, and **A*** navigation with obstacle avoidance – against a locked competition harness.
- Debugged cross-robot coordinate-frame mismatches, sim-clock vs. wall-clock timing drift, and race conditions in map-merge callbacks across independently spawned robot nodes.

Neon Fracture — Real-Time Multiplayer Browser Game

[github repo](#)

Node.js, Express.js, Socket.io, Three.js, WebGL

- Built a server-authoritative multiplayer arena shooter with a tick-based simulation loop and **WebSocket**-driven state broadcasting to keep every client in sync.
- Designed the event-driven combat/ability/scoring system end-to-end and deployed a publicly playable live build.

Production Portfolio Website — Dr. Anup Das (Agricultural Scientist)

[github repo](#)

React, TypeScript, Vite, Supabase, Framer Motion

- Delivered a client-ready portfolio site with a **Supabase**-backed admin dashboard, transcribing and structuring an 18-section biodata into a fully data-driven, dark glassmorphic UI.

Research & Professional Experience

Student Researcher — Data-Driven Core Loss Modelling in Power Magnetics

IIT Dharwad (Supervisor: Prof. Sai Ram B.)

Aug 2026 – Present

- Exploring **data-driven and machine-learning** techniques to model and predict core losses in power-magnetic components for modern power-electronic design.

Open Source Contributor — GSSoC 2026 (Open Source & AI/Agents Tracks)

GirlScript Foundation

May 2026 – Present

- Contribute agentic workflows and LLM orchestration pipelines to full-stack codebases, working alongside a global open-source contributor community.

Team Lead — The Lam Research Challenge 2026 (Team Srotas Pravaha)

Lam Research

Aug 2026 – Present

Design Team Lead

Parsec 7.0, IIT Dharwad

Aug 2026 – Present

Team Lead — Nasiko AI Hackathon (Team Mauryan Minds)

HardlyHuman

Mar 2026

Team Lead — Shaastra Smart City Challenge (S2C2 PS), Team ZeroPhase

Shaastra, IIT Madras

Dec 2025 – Jan 2026

- Led team to a **top-15 finish** among **300+ nationally competing teams**.

Leadership & Co-Curricular Activities

- Council Member**, Foc.us – Photography & Films Club, IIT Dharwad (Aug 2025 – Present); progressed Cinematographer → Video Editor → Council Member, taking on growing team-ownership at each step (Aug 2024 – Present).
- Music Director & Video Editor**, 1NOT2 Production (Sep 2025 – Present).
- Event Lead**, "Byond the Roll," & Video Editing Lead, Parsec Marketing Team, IIT Dharwad – owned event planning and execution end-to-end (Oct 2025 – Jan 2026).
- Contingent Merchandise Designer**, Inter IIT Cultural Meet 8.0 – managed concept-to-production merchandise lineup representing IIT Dharwad at IIT Kanpur (Nov – Dec 2025).
- Campus Ambassador**, Techniche, IIT Guwahati – external stakeholder outreach and on-campus promotion (May – Aug 2025).
- Marketing, Design & Content Team**, E-Summit'25, Institute Innovation Council, IIT Dharwad; **Student Volunteer**, Esummit'24.

Achievements

- Top 15 of 300+ teams**, Shaastra Smart City Challenge, IIT Madras (2026).
- 8th position (Solo Singing)** out of 46 participants from 23 IITs, Inter IIT Cultural Meet 8.0, IIT Kanpur.

Technical Skills

Languages: Python, JavaScript/TypeScript, C

AI / ML & Agentic Systems: LangChain, LangGraph, Vector Embeddings & Semantic Search (Qdrant), Multi-Agent Orchestration, Applied ML for Power Systems, Prompt Engineering, AI-Assisted Development & Debugging

Backend & Infrastructure: FastAPI, Node.js, Express.js, Docker, PostgreSQL, MongoDB, Redis, Celery, WebSockets, Kong API Gateway

Frontend & Web: React, Next.js, Three.js / WebGL, Socket.io, Tailwind CSS, Framer Motion, HTML5/CSS3

Robotics & Engineering Tools: ROS2, Gazebo, SLAM, A* Path Planning, ABB RobotStudio, MATLAB, SolidWorks

Creative & Productivity Tools: DaVinci Resolve, Adobe Lightroom, Canva, Git/GitHub

Management & Collaboration: Team Leadership, Cross-Functional Coordination, Event Management, Technical Documentation

Certifications

- Amazon AWS Solutions Architecture Job Practicum – Forage (Dec 2025)
- Operations Job Experience, Goldman Sachs – Forage (Apr 2026)
- Deloitte Australia Technology Job Externship – Forage (Jul 2025)
- Robotics & AI Industrial Training (incl. ABB RobotStudio)

Languages

English (Full Professional) | Hindi (Full Professional) | Bengali (Native / Bilingual)